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Mr. Gyu Tae Bae
153, Jeongdong-ro,
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SOUTH KOREA

Offer

Offers- No: **215718 / 5**

date: **18/11/2025**

Your technology contact

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Your Reference

Inquiry date	05/08/2025
Contact	Gyu Tae Bae
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EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 TWIN KBG for straight ball grooves (AC-joints) for the simultaneous hard milling and turning of two Inner Races of the same type

Inner Race BJ#22 - 22022-001A
Inner Race BJ#23 - 22023-001A

FINAL NEGOTIATED OFFER

Managing Directors:
Dr. Mathias Klein, Sven Hartwich, Markus Clement
Commercial Register Ulm: HRB 533796
Tax No.: 63083/68700
VAT No.: DE 813 898 840

Bank accounts:
Commerzbank AG Tübingen ˙IBAN DE09 6414 0036 0890 9103 00˙BIC COBADEFF641
Kreissparkasse Göppingen ˙IBAN DE96 6105 0000 0015 5885 93˙BIC GOPSDE6GXXX
Bayerische Landesbank München ˙IBAN DE59 7005 0000 0004 2569 67˙BIC BYLADEMMXXX

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Price does not include options (*)

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Price overview

1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 TWIN KBG (Siemens)

1.1 Basic Machine

1.1.14	1	PRICE OF MACHINE VSC 315 TWIN KBG (SIEMENS)	678.500,00 €

1.2 Additional Equipment

1.2.2	2	TOOL BLOCK FOR 1 TURNING TOOL CAPTO C6 manually clamped	32.600,00 €
1.2.4	1	CHAIN-TYPE CONVEYOR WORKPIECE LOADING, STRAIGHT, L-R (SUPPLIED BY CUSTOMER)	0,00 €
1.2.5	1	WORKPIECE STOP INNER RACE BJ#22 (SUPPLIED BY CUSTOMER)	0,00 €
1.2.6	1	WORKPIECE STOP INNER RACE BJ#23 (SUPPLIED BY CUSTOMER)	0,00 €
1.2.7	1	LIGHT BEAMS FOR LOADING CONVEYOR	1.000,00 €
1.2.8	1	LINEAR GANTRY LP5 WITH PNEUMATIC GRIPPER UNIT AND NC ROTATION AXIS	26.000,00 €
1.2.9	1	LP5 - GRIPPER JAWS INNER RACE CLAMPING DIA. 16-33mm	600,00 €
1.2.10	1	LP5 - GRIPPER JAWS INNER RACE CLAMPING DIA. 16-33mm - SPARE	600,00 €
1.2.11	1	POSITION DETECTION	2.700,00 €
1.2.12	1	NIO - CHUTE, LENGTH 1550mm, R-L / L-R	3.700,00 €
1.2.13	1	SHUTTLE ZR120TWIN R-L	36.900,00 €
1.2.14	2	WORKPIECE CARRIER RIGID INNER RACE BJ#22	3.900,00 €
1.2.15	2	WORKPIECE CARRIER RIGID INNER RACE BJ#23	3.900,00 €
1.2.16	2	AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK	2.500,00 €
1.2.17	1	CHAIN- TYPE CONVEYOR WORKPIECE UNLOADING, STRAIGHT L-R	24.400,00 €
1.2.18	1	LIGHT BEAM FOR UNLOADING CONVEYOR	950,00 €
1.2.19	1	SAFETY ENCLOSURE - SUPPLIED BY CUSTOMER	0,00 €
1.2.20	4	SAFETY SWITCH WITH CABLE FOR SAFETY ENCLOSURE	5.600,00 €
1.2.22	1	SWARF CONVEYOR, SCRAPER TYPE - REAR DISCHARGE	17.500,00 €
1.2.23	1	CHIP BOX	3.700,00 €
1.2.24	1	CENTRAL EXTRACTOR UNIT (PREPARATION)	900,00 €
1.2.26	1	EMAG - REMOTE DIAGNOSIS	1.750,00 €

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1.2.27	1	TOOL MANAGEMENT	1.500,00 €
1.2.28	1	NC-GUIDED EMERGENCY RETRACTION AT POWER FAILURE (ESR)	11.100,00 €
1.2.29	1	NETWORK DRIVE CONNECTION (SIEMENS)	2.100,00 €
1.2.30	2	INTERFACE TO EXTERNAL HANDLING SYSTEM - HARDWARE	7.800,00 €
1.2.31	1	MINI MANUAL CONTROL UNIT (SIEMENS)	4.750,00 €
1.2.32	1	CONNECTING KIT FOR MINI HAND-HELD CONTROL UNIT – SIEMENS	875,00 €
1.2.33	1	LINE MONITORING SYSTEM (LMS)	7.950,00 €
1.2.34	1	WORKSHOP-ORIENTATED PROGRAMMING AC-JOINTS (Inner Races)	6.350,00 €
1.2.36	1	TROPICAL PACKAGE	13.400,00 €
1.2.38	1	3-COLOUR STATUS LIGHT (Q-LIGHT)	1.000,00 €
1.2.39	1	PNEUMATIC PRESSURE BOOSTER	1.050,00 €
1.2.40	1	AIR DRYER FOR COMPRESSED AIR SUPPLY	6.150,00 €
1.2.41	1	AIR GUN	1.100,00 €
1.2.42	1	SPECIAL MACHINE COLOUR	3.650,00 €
1.2.43	1	DOORS FOR FLUIDIC CABINET - ADDITIONAL EQUIPMENT (DUO machine)	0,00 €
1.2.44	1	NAME PLATE FOR CHANGE OVER PROCEDURE	400,00 €
1.2.45	1	OPERATION NAME PLATE	400,00 €

1.3 Workpiece Technology

1.3.2	1	SET OF TURNING TOOLS FOR HARD TURNING	9.800,00 €
1.3.3	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR BJ#22	14.700,00 €
1.3.4	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR BJ#23	14.700,00 €
1.3.5	1	EQUIPMENT FOR BALL TRACK MILLING MACHINE	2.050,00 €
1.3.6	1	TEST BARS HSK63-C AND CAPTO C6	2.650,00 €
1.3.7	1	SETTING DIE FOR MILLING HEAD EXCHANGE	875,00 €
1.3.9	2	ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER SIZE 85 - with 1 medium duct	11.100,00 €
1.3.10	2	AIR SENSING	11.400,00 €
1.3.11	2	ADAPTER FOR SPINDLE (CENTROTEX)	23.800,00 €
1.3.12	2	MANDREL DIA. 22,0 mm - 29,0 mm (CENTROTEX)	27.800,00 €
1.3.13	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR BJ#22	9.800,00 €
1.3.14	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR BJ#22 - SPARE	9.800,00 €
1.3.15	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR BJ#23	9.800,00 €

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1.3.16	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR BJ#23 - SPARE	9.800,00 €

1.4 Service

1.4.2	1	PROJECT ENGINEERING	39.300,00 €
1.4.3	1	TECHNOLOGY DESIGN FOR INNER RACE BJ#22	6.700,00 €
1.4.4	1	TECHNOLOGY DESIGN FOR INNER RACE BJ#23	6.700,00 €
1.4.5	1	MACHINE SETUP WITH INNER RACE BJ#22	8.350,00 €
1.4.6	1	MACHINE SETUP WITH INNER RACE BJ#23	8.350,00 €
1.4.7	1	PRE-ACCEPTANCE OF THE MACHINE WITH INNER RACE BJ#22	9.750,00 €
1.4.8	1	PRE-ACCEPTANCE OF THE MACHINE WITH INNER RACE BJ#23	9.750,00 €
1.4.9	1	INSTALLATION AND DEMONSTRATION (OVERSEAS)	37.100,00 €
1.4.10	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE	5.850,00 €
1.4.11	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE	5.850,00 €
1.4.12	1	PRODUCTION SUPPORT	14.800,00 €
1.4.17	2	ADDITIONAL SET OF DOCUMENTATION	2.100,00 €
1.4.18	2	DOCUMENTATION ON USB	250,00 €
1.4.21	1	BIS CERTIFICATE	10.800,00 €
1.4.27	1	COMPLIANCE SPECIFICATION SHEET / DELIVERY CONDITIONS	12.000,00 €
1.4.29	1	OPERATING COURSE (Basic training)	10.600,00 €
1.4.30	1	MAINTENANCE (MECHANICAL AND ELECTRICAL) SERVICE COURSE (Basic training)	6.350,00 €
1.4.32	1	PACKING AND TRANSPORTATION OF TEST PIECES - OCEAN FREIGHT	5.050,00 €
1.4.33	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGES - KOR --> GER	2.100,00 €
1.4.34	1	FOB HAMBURG	22.900,00 €
1.4.35	1	UNIQUE DISCOUNT	-285.000,00 €

1.5 Machine Specification

Total: **1.015.000,00 €**

Price does not include options (*)

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1 EMAG VERTICAL TWO SPINDLE BALL TRACK MILLING AND TURNING CENTRE VSC 315 TWIN KBG (Siemens)

1.1 Basic Machine

1.1.1 1 MINERALIT (CAST GRANITE) MACHINE BED
The machine bed is manufactured from high quality Mineralit (cast granite) which guarantees superior thermal stability and excellent dampening characteristics. Mineralit has excellent vibration dampening properties which are six to eight times better than cast iron. The large distance between the high precision pre-stressed linear roller guideways of the X-axis guarantees high accuracy and high dynamics. The guideways are outside the working area and therefore require no protective covers.

1.1.2 1 ENERGY CONTAINER
Electrical cabinet, equipped to VDE 0113 and EN 60204-1 is integrated in the over-all machine frame, thus ensuring a totally transportable unit. Power supply 400 V / 50 Hz / 460 V / 60Hz (TN-C-Net).
The central supply system for lubrication of necessary components and also the service unit for compressed air supply are integrated in the energy container.

1.1.3 1 CNC-CONTROL SIEMENS
According to control specification.

1.1.4 1 HYDRAULIC UNIT
with variable speed pump drive (DvP)

1.1.5 1 COOLING UNIT (water - air)

The cooling unit regulates the heat balance of the machine.
The temperature of the coolant is adjusted to the ambient temperature with a precision control system.
The heat generated is released into the ambient air.

Technical data cooling units

1.1.6 1 MACHINE ENCLOSURE

1.1.7 2 CROSS-SLIDE
This unit is equipped with two separate axes (X and Z) per spindle. Both spindles are controlled independently by the CNC to achieve separate tool correction.
High rapid feed enable short idle times by component load/unload.
The axes drives are quick reaction frequency controlled service free AC motors which drive the slides over high precision ballscrew spindles.
The X- and Z-axis are equipped with completely enclosed linear measuring systems.
The guide of the spindle sleeve (Z-axis) is hydrostatically mounted and guarantees damping, best guidance quality and rigidity with minimal friction.
On activation of the emergency stop or in case of power failure a feed motor with integrated brake will stop the vertical axis.

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1.1.8 2 MOTOR SPINDLE
Spindle headstock unit with motor spindle
The motor spindle is a high dynamic frequency controlled service free spindle motor.
High degree of spindle rigidity due to triple bearing support over a short distance.
Precision shoulder bearings arranged in tandem (double-O). Additional bearing at spindle end for length compensation. All bearings are lifetime lubricated and thus maintenance free.
Sealing air to prevent dirt from penetrating from the outside.
The guideway system in the Z axis is equipped with hydrostatic bearings for compensation the cutting forces which guarantee excellent guideway quality by minimum friction.
Thermo-symmetrical design and integrated coolant system guarantee constantly maintained accuracy.
Positioning accuracy of spindle: +/- 0,01 angular degrees (C-axis)
Electro-hydraulic spindle locking optionally.

1.1.9 1 MILLING HEAD FOR MILLING OF BALL GROOVES (HARD MACHINING) 8500 RPM with 2 milling spindles
consisting of:
one axis, NC controlled swivelling unit (B-axis)
Swivelling housing guided by roller bearings in the side walls of the machine base.
Swivelling angle -30° to +90° (120°)
Swivelling drive by torque drive. Direct measuring system.

2 MOTOR SPINDLES HPC190 (manually tool clamping)

Max. power 40 % ED	kW	15
Torque at 40 % ED	Nm	67
Max. spindle speed	rpm	8.500
Full power at spindle speed	rpm	3.000
Spindle taper DIN 69893	HSK	C63 / F80

Fluid-cooled spindle housing. Hybrid bearing.
Lifetime lubricated. Temperature controlled.
Rotating components are precision-balanced in two planes.
Compressed air supply to tool holder pocket with max. 10 bar.
Pneumatic spindle nose cleaning device. Air seal between spindle and housing.
(warranty period 18 months)

1.1.10 1 EMAG OPERATING AND SETTING MASKS
The EMAG masks offer the setter/ operator optimal support in setting and operating the machine.
Multiple base- and monitoring functions including data input are, depending on the machine type, available.

1.1.11 1 For simple and fast programming of the machine on an appropriate ball track, the machine can be equipped with workshop-oriented programming modules. These modules are not included in the basic machine price and are described separately in the chapter „options to the CNC control“.
On the other hand, the part programs can also be completed by the operator in normal ISO code.

1.1.12 1 STANDARD COLOUR
Machine and automation - light grey to RAL 7035,
front door - grey to RAL 7046,
swarf conveyor - grey to RAL 7046,
Machine body - signal black RAL 9004.

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1.1.13	1	NOISE LEVEL An average corrected sound pressure level of max. LpAeq = 78 dB(A) to DIN 45635 is obtainded when turning with a continuous cut on a C45 steel workpiece in both longitudinal and transverse direction. If the workpiece has for example an interrupted cut due to its geometry or its clamping condition there may occur a higher sound level. For an extra cost sound deadening material can be fitted to reduce this higher sound level.	
1.1.14	1	PRICE OF MACHINE VSC 315 TWIN KBG (SIEMENS)	678.500,00 €

1.2 Additional Equipment

1.2.1		**TOOLING SYSTEM**	
1.2.2	2	TOOL BLOCK FOR 1 TURNING TOOL CAPTO C6 manually clamped consisting of: toolholder with clamping insert C6 With internal air blast feeding to the tools. The coolant system guarantees constantly maintained accuracy.	32.600,00 €
1.2.3		**COMPONENT STORAGE AND TRANSPORT SYSTEM**	
1.2.4	1	CHAIN-TYPE CONVEYOR WORKPIECE LOADING, STRAIGHT, L-R (SUPPLIED BY CUSTOMER) WORKPIECE LOADING - LEFT, by motorised chain-type conveyor Basic design with posts. Workpiece transport lining up on a circulating link chain. Side guides synchronously adjustable via slot, with a clamping lever (ahead curve). Side guides fixed (behind curve) 1x pneumatic stopper / separator ahead curve, adjustable. 2x light beams. Electric / pneumatic installation on plug distributor. Conveyor chain width 82,5 mm Workpiece diameter 35 mm - 160 mm Workpiece weight max. 7,0 kg	
1.2.5	1	WORKPIECE STOP INNER RACE BJ#22 (SUPPLIED BY CUSTOMER) Stop for Inner Race Designed for one workpiece diameter	
1.2.6	1	WORKPIECE STOP INNER RACE BJ#23 (SUPPLIED BY CUSTOMER) Stop for Inner Race Designed for one workpiece diameter	
1.2.7	1	LIGHT BEAMS FOR LOADING CONVEYOR 1x storage position free / 1x loading conveyor full For external loading	1.000,00 €

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1.2.8	1	LINEAR GANTRY LP5 WITH PNEUMATIC GRIPPER UNIT AND NC ROTATION AXIS 2-axis linear gantry driven by a NC servo motor. Direct connection to the machine. Energy supply on all axes via the energy supply chain. Rotation unit driven by a servomotor with a rotary feedthrough. Workpiece transport with a pneumatic 3-finger centric gripper and NC rotation axis (360°) X-axis stroke 800 mm, Z-axis stroke 600 mm Max. weight of workpiece 7 kg Plus workpiece-specific gripper jaws Prepared for the attachment of an automatic workpiece alignment station for angle-oriented positioning in the ball tracks (hard machining)	26.000,00 €
1.2.9	1	LP5 - GRIPPER JAWS INNER RACE CLAMPING DIA. 16-33mm 1 set of gripper jaws for 3-finger centric part gripper	600,00 €
1.2.10	1	LP5 - GRIPPER JAWS INNER RACE CLAMPING DIA. 16-33mm - SPARE 1 set of gripper jaws for 3-finger centric part gripper	600,00 €
1.2.11	1	POSITION DETECTION Position detection of the workpiece via Keyence laser sensor, mounted on the gantry frame	2.700,00 €
1.2.12	1	NIO - CHUTE, LENGTH 1550mm, R-L / L-R Unloading, flow direction R-L / L-R Design as a chute with structural sheet metal For non-oriented or wrong workpieces, consisting of: Direction to the side → right / left 1x light beam Length: approx. 1550 mm Accumulation range: approx. 1350 mm	3.700,00 €
1.2.13	1	SHUTTLE ZR120TWIN R-L for the workpiece transport through the machine. With two basic pallets for the unmachined components. With two cardanic, spring-loaded pick-up stations, for the positive location of workpieces in the chuck The components are automatically positioned for pick-up by the pick-up spindle. Various component diameters and heights are defined in the NC program. Consisting of: 2x spring loaded pickup stations, including detection and emergency stop . Driver unit for workpiece receptor. Slide-/brake plate topped with spring steel. 1x light beam Driven by servomotor CV joints are conveyed vertically, with the CV joint bell pointing downward.	36.900,00 €
1.2.14	2	WORKPIECE CARRIER RIGID INNER RACE BJ#22 Hard machining for the location of inner races. Designed for one workpiece.	3.900,00 €
1.2.15	2	WORKPIECE CARRIER RIGID INNER RACE BJ#23 Hard machining for the location of inner races. Designed for one workpiece.	3.900,00 €

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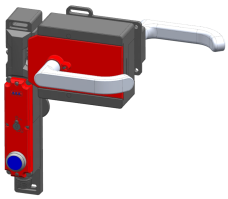
1.2.16	2	AIR BLAST DIRECTED AT THE PICK-UP UNIT AND CLAMPING CHUCK For the cleaning of the pick-up unit and the clamping chuck by air blast. The air is processed by a service unit and directed at the component through nozzles.	2.500,00 €
1.2.17	1	CHAIN- TYPE CONVEYOR WORKPIECE UNLOADING, STRAIGHT L-R WORKPIECE UNLOADING - RIGHT, by motorised chain-type conveyor Basic design with posts. Workpiece transport lining up on a circulating link chain. Side guides synchronously adjustable via slot. Light beams. Electric / pneumatic installation on plug distributor. Conveyor chain width 82,5 mm Workpiece diameter 35 mm - 160 mm Workpiece weight max. 7,0 kg	24.400,00 €
1.2.18	1	LIGHT BEAM FOR UNLOADING CONVEYOR 1 x workpiece at the unloading position For external unloading	950,00 €
1.2.19	1	SAFETY ENCLOSURE - SUPPLIED BY CUSTOMER **Please note:** **- EMAG will supply an incomplete / unsafe machine.** **- The customer is responsible for the safety of the system (machine and periphery).**	
1.2.20	4	SAFETY SWITCH WITH CABLE FOR SAFETY ENCLOSURE for customer supplied safety enclosure consisting of: - cable - closing mechanism	5.600,00 €



Price does not include options (*)

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1.2.21		**COOLANT AND SWARF DISPOSAL SYSTEM**	
1.2.22	1	SWARF CONVEYOR, SCRAPER TYPE - REAR DISCHARGE Swarf exit height 1100 mm. Located to the rear of the machine. TEM drive motor with overload protection via torque clutch and microswitch Usage: for short steel swarf (10-60mm).	17.500,00 €
1.2.23	1	CHIP BOX Compact skip container 1000 l - incl. wheel set (load capacity 2500 kg)	3.700,00 €
1.2.24	1	CENTRAL EXTRACTOR UNIT (PREPARATION) Connector for central extractor unit, mounted at the machining area. Diameter 150 mm. Required extractor capacity approx. 1000 m^3/h Supply of a suitable extractor unit, for setup and pre acceptance of the machine at manufacturer's plant, by the customer or of EMAG at extra costs.	900,00 €

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1.2.25 **OPTIONAL EQUIPMENT FOR CNC CONTROL**

1.2.26 1 EMAG REMOTE DIAGNOSIS 1.750,00 €

Remote access to the control system of the machine / automation for detailed error diagnosis and the possibility to support the customer in case of queries or problems. Remote access takes place via a secure VPN connection with a router including firewall (the VPN connection is activated via a key switch).

Preconditions:

- Linkage with user network necessary via DHCP
- Internet connection of the machine (only during the remote diagnosis session) to the public IP address of EMAG
- The following IP segments may not be used
  - 192.168.0.0/24
  - 192.168.1.0/24
  - 192.168.143.0/24
  - 192.168.144.0/24
  - 192.168.214.0/24
  - 192.168.216.0/24
  - 172.30.0.0/16

Your advantages:

- Network separation through the use of a router
  (Isolation of the machine through separate IP segments, thereby increasing the security level and reducing attack surfaces)
- Simple handling for the customer
- Quick, cost-effective support, resulting in reduced downtimes.
- Up-to-date problem solving without lengthy delays.
- Program extensions can be realized at a reduced cost.
- "On-the-spot" co-operation between specialist and operating/service personnel allows for complex errors to be diagnosed, analysed and eliminated quickly.

Availability:

The service is available at the local service times for the following countries:
Europe (excluding Russia), China and USA.
For all other countries the service is linked to the service times of the CET.
Free of charge during the warranty period, thereafter at the current hourly service rates.

1.2.27 1 TOOL MANAGEMENT 1.500,00 €

The functions of the “tool management” ensure that the correct tool is selected on the machine at all times and that the data assigned to the tool is up-to-date.
The tool management also has the following advantages:

- Avoidance of rejects by monitoring the use of tools via an empirically determined number of pieces (tool life), the tool life time or, in the case of grinding wheels and dressers, usually via wear.
- Avoidance of machine downtimes by considering replacement tools.
- An optimal providing of the tools for the tool change.

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1.2.28	1	NC-GUIDED EMERGENCY RETRACTION AT POWER FAILURE (ESR) Interpolated retraction of axis in case of power failure, voltage fluctuation and drive failure. Electric components (condenser modul) are used which maintain the voltage for the duration of the retreat and use the energy for the nc-guided retract of the axes. (strongly recommended for unstable power supply, expensive tools or workpieces)	11.100,00 €

1.2.29	1	NETWORK DRIVE CONNECTION (SIEMENS)
		Linking the machine control with your local network via a router, so that network drives are available in the HMI.
		NC-programs or any data can be stored or loaded from there.
		
		<u>Customer-specific requirements / restrictions for network integration:</u>
		• Network cable into the electrical cabinet
		• Server- and sharename of the network resource
		• User name and password for the automatic preparation of the connection
		• Operation with DHCP server
		• The following IP segments may not be used
		  • 192.168.0.0/24
		  • 192.168.1.0/24
		  • 192.168.143.0/24
		  • 192.168.144.0/24
		  • 192.168.214.0/24
		  • 192.168.216.0/24
		  • 172.30.0.0/16
		
		When providing server and directory names it has to be taken into account that the control features an 'English language operating system' that cannot handle the German umlauts ä, ö and ü.
		When installing of security update, the company Siemens restrictions are to consider.
		If in the domain automatically software is installed on all computers, so the functionality is no longer guaranteed after restart!
			2.100,00 €

1.2.30	2	INTERFACE TO EXTERNAL HANDLING SYSTEM - HARDWARE Safety signal interface via HAN24E plug on electrical cabinet with 8 potential-free inputs and 8 potential-free outputs for the change of safety signals "emergency-stop" and "access door closed" Handling-signal interface via HAN16E plug on electrical cabinet Deviations of the standard interface on request at additional costs. (1.000,00 € / signal)	7.800,00 €
1.2.31	1	MINI MANUAL CONTROL UNIT (SIEMENS) For operating a combined positioning and transfer unit and/or a linear gantry loader. In the setting up mode of the automation components: the shuttle, the vertical and horizontal axis of the gantry, also the swivelling and gripping function of the part gripper can be manually operated with the manual control unit. If the doors of the safety enclosure are open, the acknowledgment key must be pressed on the manual control unit.	4.750,00 €
1.2.32	1	CONNECTING KIT FOR MINI HAND-HELD CONTROL UNIT – SIEMENS	875,00 €

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1.2.33	1	LINE MONITORING SYSTEM (LMS) Same design and function as the supplied machines SN- 117021 and 117022. Adaptations required due to new Sinumerik One control system.	7.950,00 €
1.2.34	1	WORKSHOP-ORIENTATED PROGRAMMING AC-JOINTS (Inner Races) For Inner Races (straight grooves). Suitable for Inner Race BJ#22 and BJ#23. For quick and easy programming parameters for componentrelated dimensions are adapted. Same process for tool-offset.	6.350,00 €

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1.2.35 **COUNTRY SPECIFIC EQUIPMENT**

1.2.36 1 TROPICAL PACKAGE 13.400,00 €

for max. 48ºC ambient temperature on machines with "water / air" cooling unit
for max. 52ºC ambient temperature on machines with "water / water" cooling unit
at least 10°C with grease lubrication
at least 5°C with oil lubrication
at max. 90% humidity

Depending on the machine type, the following components are installed:
- Control cabinet heater
- more powerful cooling unit

more powerful / additional control cabinet cooling unit

1.2.37 **OTHER AUXILIARY EQUIPMENT**

1.2.38 1 3-COLOUR STATUS LIGHT (Q-LIGHT) 1.000,00 €
mounted on machine. Colour code:
red: machine stopped
yellow: warning -tool to be changed,
-lubricant running low, etc.
green: machine in automatic cycle.

1.2.39 1 PNEUMATIC PRESSURE BOOSTER 1.050,00 €
to increase air pressure from 4 bar to 6 bar.
(only for part grippers)
Maker: Festo
Type: DPA-63-10

1.2.40 1 AIR DRYER FOR COMPRESSED AIR SUPPLY 6.150,00 €
Absorption air dryer for dry and clean air.
Manufacturer: Festo
Type: PDAD-51-G3/8
incl. combined maintenance unit. Only sealing air is dried. Blown air is not dried.

1.2.41 1 AIR GUN 1.100,00 €
Tube in transparent blue

1.2.42 1 SPECIAL MACHINE COLOUR 3.650,00 €
Complete machine according to customers colour specification.

1.2.43 1 DOORS FOR FLUIDIC CABINET - ADDITIONAL EQUIPMENT (DUO machine)
The fluidic cabinets of the machine (pneumatic + hydraulic) can be provided with doors and closed.
4 doors equipped with Makrolon windows.

1.2.44 1 NAME PLATE FOR CHANGE OVER PROCEDURE 400,00 €
Size A3
Made of aluminium.

1.2.45 1 OPERATION NAME PLATE 400,00 €
With PVC (400L x 140H x 8T),
Description : OP No. and OP Name
Made of aluminium.

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1.3 Workpiece Technology

1.3.1 TOOL EQUIPMENT

1.3.2	1	SET OF TURNING TOOLS FOR HARD TURNING consisting of: - 4 turning tools Capto C6 - 40 cbn cutting inserts	9.800,00 €
1.3.3	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR BJ#22 Head system with 4 brazed cbn cutting inserts consisting of: - 4 basic tool holder HSK63-C - 30 cutting heads	14.700,00 €
1.3.4	1	SET OF MILLING TOOLS FOR HARD MILLING OF IR BJ#23 Head system with 4 brazed cbn cutting inserts consisting of: - 4 basic tool holder HSK63-C - 30 cutting heads	14.700,00 €
1.3.5	1	EQUIPMENT FOR BALL TRACK MILLING MACHINE consisting of: - 1 cone wiper - 1 torque wrench with insert - 2 plugs HSK63 - 1 ball plug gauge HSK63	2.050,00 €
1.3.6	1	TEST BARS HSK63-C AND CAPTO C6 for checking the machine geometry Consisting of: 1x aluminum case 1x foam insert 1x test bar HSK63-C 1x test bar Capto C6	2.650,00 €
1.3.7	1	SETTING DIE FOR MILLING HEAD EXCHANGE consisting of : 1x fixture for milling head change for mounting on a bench.	875,00 €

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1.3.8		**CHUCKING SYSTEM**	
1.3.9	2	ROTATING SOLID CENTRE HYDRAULIC CHUCKING CYLINDER SIZE 85 - with 1 medium duct Max. drawbar force 3.325 daN at 70 bar. The clamping pressure is manually infinitely adjustable. The clamping cylinder stroke and the clamping pressure are controlled by a linear measuring system and by a pressure switch. Coolant or air is supplied through the main spindle. Incl. connecting parts with standard threaded connection and control dimension.	11.100,00 €
1.3.10	2	AIR SENSING The air sensing detects/monitors the workpiece positon on the clamping device.	11.400,00 €
1.3.11	2	ADAPTER FOR SPINDLE (CENTROTEX) for quick change of the clamping mandrels / collet chucks Prepared for air sensing. The adaption of the clamping device to the draw bar is via a bayonet.	23.800,00 €
1.3.12	2	MANDREL DIA. 22,0 mm - 29,0 mm (CENTROTEX) For clamping of Inner Races. Installation by CentroTex-interface for quick change of the mandrel (spindel adapter is required). Round clamping geometry. Clamping range: dia. 22,0 mm - 29,0 mm Prepared for air sensing. Without workpiece-specifically interchangeable parts.	27.800,00 €
1.3.13	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR BJ#22 Laid out for one inner race. Consisting of collet, part stop and in total 1x control ring.	9.800,00 €
1.3.14	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR BJ#22 - SPARE Laid out for one inner race. Consisting of collet, part stop and in total 1x control ring.	9.800,00 €
1.3.15	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR BJ#23 Laid out for one inner race. Consisting of collet, part stop and in total 1x control ring.	9.800,00 €
1.3.16	2	SET OF INTERCHANGEABLE PARTS (MANDREL) - IR BJ#23 - SPARE Laid out for one inner race. Consisting of collet, part stop and in total 1x control ring.	9.800,00 €

1.4 Service

1.4.1	1	Please note that no additional services are contained in the previously specified prices. We would be pleased to carry out the following services and would quote according to your wishes. - Preparation of tooling layouts and NC programs - Set-up machine for component trials on your specific workpiece in our works - Machine approval in our works by your engineers. - Final acceptance at customer's plant with a specific workpiece

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1.4.2	1	PROJECT ENGINEERING Project engineering and coordination with suppliers and customers. Engineering investigations (machine set up, with central automation, central exhaust, chip disposal) Customization of interfaces with the suppliers.	39.300,00 €
1.4.3	1	TECHNOLOGY DESIGN FOR INNER RACE BJ#22 Design of the complete tooling- and clamping system for one workpiece. Consisting of: - selection of manufacturer (in coordination with the customer) - requests for tools and clamping devices - collision analysis - design of the tooling layout - approval and release process between EMAG and customer - approval and release process between EMAG and suppliers (tools & clamping devices) - responsibility of the complete tooling and clamping equipment - creation of the machining program (if provided by the customer, checking the travel paths, cycles and adjustments to the control) If the tools and clamping devices are supplied by the customer, the design is carried out by EMAG based on the EMAG cycle time calculation in coordination with the customer.	6.700,00 €
1.4.4	1	TECHNOLOGY DESIGN FOR INNER RACE BJ#23 Design of the complete tooling- and clamping system for one workpiece. Consisting of: - selection of manufacturer (in coordination with the customer) - requests for tools and clamping devices - collision analysis - design of the tooling layout - approval and release process between EMAG and customer - approval and release process between EMAG and suppliers (tools & clamping devices) - responsibility of the complete tooling and clamping equipment - creation of the machining program (if provided by the customer, checking the travel paths, cycles and adjustments to the control) If the tools and clamping devices are supplied by the customer, the design is carried out by EMAG based on the EMAG cycle time calculation in coordination with the customer.	6.700,00 €
1.4.5	1	MACHINE SETUP WITH INNER RACE BJ#22 for one workpiece. Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	8.350,00 €
1.4.6	1	MACHINE SETUP WITH INNER RACE BJ#23 for one workpiece. Optimisation of interfaces and system software. Fitment and test run of workpiece-specific clamping devices and tooling equipment. Test run of NC program, ready for customer's machine acceptance. Tool life optimisation not included.	8.350,00 €

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1.4.7	1	PRE-ACCEPTANCE OF THE MACHINE WITH INNER RACE BJ#22 Specific workpiece: **Inner Race BJ#22 - 22022-001A** including measuring of workpieces at EMAG (3D measuring machine, roundness measuring device, etc.). One workpiece in one clamping operation. Max. 30 components/workpiece spindle. Proof of machine capability Cm as agreed in the EMAG quality data sheet.	9.750,00 €
1.4.8	1	PRE-ACCEPTANCE OF THE MACHINE WITH INNER RACE BJ#23 Specific workpiece: **Inner Race BJ#23 - 22023-001A** including measuring of workpieces at EMAG (3D measuring machine, roundness measuring device, etc.). One workpiece in one clamping operation. Max. 30 components/workpiece spindle. Proof of machine capability Cm as agreed in the EMAG quality data sheet.	9.750,00 €
1.4.9	1	INSTALLATION AND DEMONSTRATION (OVERSEAS) in your works The following services are included: - Machine installation and demonstration - Machine approval and hand-over - Talk-through of the machine documentation and service instruction - Operator instruction - Travel costs Total of 15 man days Not included are: - Machine set-up - Machine operation for production - Additional service Additional service engineer costs for time in excess of that quoted will be charged extra.	37.100,00 €
1.4.10	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE Specific workpiece: **Inner Race BJ#22** Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces/workpiece spindle. Acceptance criteria at the manufacture's plant as agreed in the EMAG quality data sheet.	5.850,00 €
1.4.11	1	FINAL ACCEPTANCE AT CUSTOMER'S PLANT WITH SPECIFIC WORKPIECE Specific workpiece: **Inner Race BJ#23** Gauging of the workpieces with measuring equipment supplied by customer. Uninterrupted machining of 30 workpieces/workpiece spindle. Acceptance criteria at the manufacture's plant as agreed in the EMAG quality data sheet.	5.850,00 €
1.4.12	1	PRODUCTION SUPPORT 1 week (5 working days, monday - friday) production support at customers plant.	14.800,00 €

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1.4.13 1 INSTALLATION POINT

The installation point has to be prepared in accordance with the documentation supplied. We take it for granted that all conditions are met before our installation crew arrives.
Where waiting times are incurred due to the conditions not having been met, these will be charged extra. During installation and commissioning our engineers have to be provided, on request, with the support of skilled customer personnel and similar assistance. This also applies to the operation of cranes, handling equipment and other aids.

1.4.14 1 MACHINE LIFTING EQUIPMENT

EMAG will provide a lifting equipment to lift the machine by crane, free of charge. However, should the customer not return the equipment to EMAG within 4 weeks, freight and charges prepaid, a separate invoice of EUR 1.500,00 will be issued.

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1.4.15 **DOCUMENTATION**

1.4.16 1 DOCUMENTATION (Siemens)

OPERATING INSTRUCTIONS
1 complete set of operating instructions on paper
EU countries in the official language of the country; other countries on national regulations

Manuals for the machine:
- Introduction
- Safety instructions
- Product description
- Transport and installation
- Start-up
- Production
- Preventive maintenance

Manuals from the manufacturer of the control unit:
Siemens Sinumerik
- User manual
- Diagnostics manual
- Programming manual

Machine diagrams:
- Machine layout
- Transport diagram
- Machine data sheet
- Drive data sheet
- Connection data sheet for fluidics
- Connection data sheet for electrics

List of spare parts and wearing parts

Operating instructions from third parties

Documentation for clamping devices and tools

TECHNICAL DATA
1 set of technical data on paper, German or English
- Electrical circuit diagram with list of el. devices
- Fluid system diagram with list of el. devices
- Drawings of subassemblies with relevance for spare parts
- List of components with relevance for spare parts and wearing parts in relation to the drawings
- Technical data, test reports, declarations of incorporation from third parties

Other languages available on request for an additional charge; third party documentation excepted

1.4.17 2 ADDITIONAL SET OF DOCUMENTATION 2.100,00 €
1 additional set of the documentation described above

1.4.18 2 DOCUMENTATION ON USB 250,00 €
1 complete set of documentation on USB

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1.4.19 **CE-MARKING**

1.4.20 1 WITHOUT CE MARKING / WITHOUT DECLARATION OF CONFORMITY (non-EU country, partly completed machinery)
The machinery is always designed and constructed in accordance with the applicable harmonized standards and safety instructions.
By request of the customer, the machinery is constructed as partly completed machinery (e.g. in the workpiece infeed and outfeed area) and is intended exclusively for incorporation into or assembly with other machinery or equipment.
The owner-user is responsible for securing these areas. The machinery may not be put into service until the owner-user has declared that the final machinery into which this partly completed machinery has been incorporated is in conformity with the regional provisions.

1.4.21 1 BIS CERTIFICATE 10.800,00 €
Certification by the Bureau of Indian Standards.
Application, testing and issuing an authorisation that allows the product to be labelled.

Please note that this item is excluded from negotation and is not discounable.

Price does not include options (*)

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VAT No.: DE 813 898 840

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1.4.22 **LANGUAGES**

1.4.23 1 LANGUAGE OF DOCUMENTATION
Content as described above
1 complete set of operating instructions
EU countries in the official language of the country; other countries on national regulations

Language for 1st set (costs included in machine price) **
Desired language: **English***

Language for additional set (additional price)
Desired language: **English***

Possible languages (*to be defind with the customer)**:**

Bulgarian	Portuguese
Danish	Romanian
German	Swedish
English	Slovakian
Estonian	Slovenian
Finnish	Spanish
French	Czech
Greek	Hungarian
Italian	
Croatian	Chinese (Simplified)
Latvian	Japanese
Lithuanian	Korean
Dutch	Russian**
Polish	Turkish

**Additional price, note the description for documentation in Russian

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1.4.24 1 LABEL LANGUAGE
The machine labels are generated in 2 languages (bilingual)
First language is always the official language of the destination country. Second language is English if possible.

1st Language
Desired language: **English***

2nd Language
Desired language: **German**

1st Possible languages (official language of the destination country**)***to be defind with the customer**:**

Bulgarian	Portuguese
Danish	Romanian
German	Swedish
English	Slovakian
Estonian	Slovenian
Finnish	Spanish
French	Czech
Greek	Hungarian
Italian	
Croatian	Chinese (Simplified)
Latvian	Japanese
Lithuanian	Korean
Dutch	Russian
Polish	Turkish

1.4.25 1 SCREEN LANGUAGE (Siemens)
The screen language refers to the language of the system software and EMAG specific applications.
Applications from third-party suppliers and EMAG CNC cycles are not included.

1st Language
Desired language: **German**

2nd Language
Desired language: **English**

Price does not include options (*****)

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1.4.26 **CUSTOMER SPECIFICATION**

1.4.27 1 COMPLIANCE SPECIFICATION SHEET / DELIVERY CONDITIONS 12.000,00 €

Design of mechanical-, electrical- and fluid equipment according to " ENG_RFQ_for inner Race_WIA India Volvo GPA_WGT-M3-0019-001 "
Due to our design we are unable to comply with the following items of your specification sheet, meaning we must apply for an exceptional permission.

Item	Customer requirement	EMAG commitment
Basic Condition	• AC 415V (+/-15%) x3 phase x 50Hz • Surrounding temperature -5°C - 50°C	• AC 415V (+/-10%) x3 phase x 50 Hz like in your specifications item Electric 1. Main Circuit. • With tropical package surrounding temperature between 10°C and 48°C
Noise level	less than 75 dB (A)	up to 80dB (A)
General specification of Hyundai WIA	Keep the "General specification of Hyundai WIA"	This document is not part of the inquiry and therefore it is not considered
Workpiece quality	Different requirements on pages 4 and 5 of the RFQ. Due to this, we unfortunately have to decline the requirements.	Regarding quality commitments please see attached quality data sheets. Only the EMAG quality data sheets are valid.
Main Circuit	Solenoid valve: 115V AC 50Hz, Lamp Type, Communication Type	24V DC with LED
CNC & PLC	• Voltage 240V • Memory USB Card • Memory EPROM	• 400V AC / 24V DC • Memory USB-Stick • No longer available from Siemens.
Safety fence	Need	Supplied by customer
CNC & PLC	Network interface - LAN	USB
Wiring	Remain the 30% marginal space and do not wire it hose together	max. 10% possible
Automation height	FL+950mm	FL+1030mm / FL+1050mm - Same as delivered with last MC's
Items for the capability (pre- and final acceptance)	Ball Track Milling: Groove Dia, Center, Ball Diameter	Ball diameter is no item for evaluation (Cm)

Price does not include options (*)

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1.4.28	1	TRAINING	
1.4.29	1	OPERATING COURSE (Basic training) For two participants in your works. Directly after installation and demonstration. Matched to the machine design (modular customized) Training course duration: 5 days / shift (shift monday - friday). Language: Marathir language Training documentation (3x) language: English	10.600,00 €
1.4.30	1	MAINTENANCE (MECHANICAL AND ELECTRICAL) SERVICE COURSE (Basic training) For two participants in your works. Matched to the machine design (modular customized) Training course duration: 3 days / shift (shift monday - friday). Language: Marathir language Training documentation (3x) language: English	6.350,00 €
1.4.31		**TRANSPORT COSTS**	
1.4.32	1	PACKING AND TRANSPORTATION OF TEST PIECES - OCEAN FREIGHT Transport (ocean freight) from customers plant in Southern Korea to EMAG Maschinenfabrik in Salach. - 2x 250 test pieces of each acceptance part In total 2 pallets (each pallet with max. size and weight: 1200x800x1000mm; max. weight = 600kg) **The customer has to take care of a transport-safe packaging in each case.** **The workpieces will be send together with the machine back to Hyundai WIA India.** **Please note that these prices are updated daily. They are subject to change and EMAG Maschinenfabrik reserves the right to adjust the prices accordingly (due to the current situation in the red sea). Additional costs and/or reduced costs will be passed on to the customer Hyundai WIA.**	5.050,00 €
1.4.33	1	PACKING AND TRANSPORTATION OF INSPECTION GAUGES - KOR --> GER Transport from customers plant in Korea to EMAG Salach Germany. 1x euro pallet 120x80x80 (Ocean freight, max. 150kg) **The customer has to take care of a transport-safe packaging in each case.** **It must be ensured that EMAG can reuse the packaging material.**	2.100,00 €
1.4.34	1	FOB HAMBURG from Salach, Incoterms 2020	22.900,00 €
1.4.35	1	UNIQUE DISCOUNT Unique special project discount.	-285.000,00 €

Price does not include options (*)

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1.5 Machine Specification

1.5.1 1 TECHNICAL MACHINE DATA VSC 315 TWIN KBG subject to change without notice

WORKING AREA		
Chuck diameter	mm	254
Swing diameter	mm	260
X-stroke	mm	935
Z-stroke	mm	315
WORKPIECE DIMENSIONS		
Workpiece length, max.	mm	280*
Workpiece O/D, max.	mm	160*
Ball section dia., max.	mm	30
(*Longer workpieces and bigger diameter on request)		
MAIN SPINDLE	Qty	2
Spindle flange acc. to DIN ISO 702-1	size	8
Spindle front bearing diameter	mm	140
max. spindle speed	rpm	3000
Spindle distance	mm	500
MAIN DRIVE	Qty	2
AC-Synchronous motor 40/100 % ED	kW	28,5/27,5
Max. power	kW	28,5
Full power at spindle speed	rpm	700
Torque at 40/100 % ED	Nm	590/375
Max. torque at	Nm	590
FEED DRIVE		
Rapid traverse feed, X	m/min	45
Rapid traverse feed, Z	m/min	30
Feed force, X/Z	kN	11
Ball screw diameter X/Z	mm	50/40
MILLING SPINDLE		
AC-motor	kW	15
Max. spindle speed	rpm	8500
ELECTRICAL EQUIPMENT according to VDE 0113		
Supply voltage	V	400
Control circuit voltage - DC	V	24
- AC	V	230
Frequency	Hz	50
Total connected load	kVA	70
Input fuse	A	160
REQUIRED COMPRESSED AIR QUALITY		
according to DIN ISO 8573-1, class 4	bar	5-6
DIMENSIONS AND WEIGHT		
Length/Width/Height	mm	4980/2800/3000
Weight	kg approx.	13000

Price does not include options (*)

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1.5.2 1 OPERATOR PANEL 19" MULTITOUCH (Siemens)
consisting of IFP1900 and Control panel (MCP1900)

Price does not include options (*)

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1.5.3 1 CNC CONTROL SIEMENS SINUMERIK ONE
with integrated PLC-1500F and IPC

- Operator software Sinumerik Operate
- Screen language and on-screen messages in German / English / in official EU language, more languages available as an option
- CNC program and data memory 10 MB, optionally expendable in 2 MB steps (max. 28 MB)
- USB-interface (front) for memory-stick
- Data backup on external media (USB stick integrated into IPC) according to EMAG standard
- dataGrabber (Comara) – Capable of recording process data for diagnostics. No active evaluation possible on the control, storage of the data takes place in a ring buffer
- Siemens standard cycles are supported (technology-dependent, additional options may be required for certain applications)

EMAG ADD-ON FUNCTIONS

- Machine-specific documentation of functions and parameters is available directly on the control unit.
- Automatic return to the home position. The process is continued after an interruption without much effort.
- Reduced rapid traverse speed during the first run. This reduces or even eliminates follow-up costs that may arise from incorrect programming or operating errors
- Automatic filling and emptying of the entire machine
- “Optional Stop” (M01) allows you to make a programmed stop or interrupt production after finishing a part at a desired point
- Single cycle: Allows one part to be transferred and machined for setup and tryout purposes

Structured work piece programming:

- Division of individual programs into subprograms, e.g. the program for the machining contains only the technology program and not the loading part. The load-/unloading program has individual parameters. Division into machine and work piece specific parameters
- Programmable chuck with length measuring system in the pull rod. Different clamping situations (Unclamped, Clamped and Over stroke) can be changed by parameter assignment
- Recording and displaying of the most important production data: work piece counter, shift counter, cycle time, work piece status, operating hours
- Production of definable batch sizes. The machine stops after the required quantity is reached. Available for individual machines; for interlinked systems, this option must be checked on a case-by-case basis
- Parameterizable swiveling of the turret to eject chips (technology-dependent)
- Rinsing of clamping devices after a programmable number of parts (depends on technology)

Price does not include options (*****)

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- Number and positions of the workpiece carriers can be defined for closed automation systems
- Lifecycle screen that displays tool wear and tool life (Siemens tool management)

Total: **1.015.000,00 €**

Price does not include options (*)

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Commercial Conditions

CONFIDENTIALITY
Any commercial and operational written or oral information or data relating to this quotation / order, such as drawings, specifications, recommended machining data and clamping methods, etc., shall not be divulged to any third party (referring, in particular, but not exclusively, to competitors, for the purpose of bid comparisons) and are to be used solely within the meaning of this quotation / order.
The recipient of this quotation / order expressly accepts these obligations.

MACHINING
The determined cycle times result from values, which are based on data and experiences of acknowledged cutting material manufacturers. Changing rough part dimensions, tolerances, material as well as the material hardness are only possible during building and testing of the machine. Changes will only be accepted in written form and in mutual agreement.

PROVISION OF EQUIPMENT AND MATERIALS BY CUSTOMER
All tooling, work holding equipment, parts, etc. are to be supplied with DDP supplying plant EMAG Incoterms 2020. We wish to point out that we accept no responsibility for the functional capability and working order of any equipment provided.
The functional capability of the provided equipment is the responsibility of the buyer.
In the event of problems, we assure you of our constructive collaboration. Any additional costs arising from modification, optimization, etc. will be charged to the customer. Furthermore, we do not guarantee any quoted machining times in such cases. If items supplied by the customer are included, the user documentation, declarations of conformity, and declarations of incorporation of the items supplied by the customer have to be provided to EMAG. The item supplied by the customer has to fulfill the requirements of the Machinery Directive 2006/42/EC.
When orders cannot be carried out on schedule, caused by material supplied by the customer or if technical problems should occur, the machine will be invoiced on the originally agreed delivery date, when the relevant payment will become due.
The agreed upon date of delivery date may be postponed to the time to achieve full operability, which shall not result in any event of default of the manufacturer.

DELIVERY PERIOD
Approx. 11 months FOB German Seaport. The delivery period is not binding and can change between the submission of quotation and placement of order.
Please discuss the subject of delivery with us at time of order placement and we will do everything possible to meet your requirements. Changes regarding the component/s or the machines or the order content after order placement may result in changes in price and delivery.
Compliance of delivery date requires timely down and / or partial payment by the customer within the prescribed payment terms. Adherence to the delivery date is subject to timely down payment by the customer within the stipulated periods. If the receipt of the down payment is postponed, the delivery date and the start of a possible penalty for late delivery shall be postponed accordingly.

DELIVERY BOTTLENECKS
If we are unable to meet delivery dates (non-availability of the service) for reasons for which we are not responsible (particular "Force Majeure"), we reserve the right to postpone the delivery date to a point in time at which the service is possible for us again. This applies for a maximum period of twelve months beyond the agreed delivery time. If the unavailability of the service will not end foreseeably, we reserve the right to withdraw from the contract in whole or in part.
A case of non-availability of the service also applies in particular to non-delivery by our suppliers (e.g. disrupted supply chains worldwide), unless EMAG is responsible for this circumstance through intentional or grossly negligent behavior.

We assure you of our endeavor to serve every order in a timely manner.

CONFORMITY WITH CUSTOMERS` REGULATIONS
has not been taken into account. Where unavoidable deviations from such regulations prove necessary, specific authorization will be requested.

Price does not include options (*****)

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PRICES
FCA (Free Carrier, Incoterms 2020). Packing, insurance and transport not included.
The payments are to be made without deduction. With delayed payment the relevant laws apply.

VALIDITY OF PRICES
The prices quoted are valid for 60 days from the date of quotation. Orders placed after this period would be subject to a price increase of 1 % per month to cover increases in labour and material costs.
This quotation will replace all previous versions.

Price does not include options (*)

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TERMS OF PAYMENT
30 % advance payment by T/T
70 % of invoice value against irrevocable Letter of Credit payable as follows:
60 % of L/C-value at sight against shipping documents.
10% of L/C-value at sight against copy of final acceptance protocol

All L/C-charges are for buyer's account. L/C must be opened latest 30 days after order confirmation.

RETENTION OF TITLE
The machine will remain the property of the builder until complete payment is received from the buyer. Upon placement of order the extended retention of title according to clause 48 of our General Conditions for Sales and Deliveries shall be deemed as agreed.

LIMITATION OF LIABILITY
Liability of the contractor, for whatever legal reason, is excluded. This does not apply in cases of intent or gross negligence, in cases of injury to body, life or health, as well as in cases of product liability or liability from other mandatory statutory provisions.

PERIOD OF LIABILITY
Any rights and claims of the buyer caused on defects shall be subject to a limitation period of twelve months.
The limitation period starts with the final acceptance, but no later than 3 months from delivery, and does not apply to wear and tear parts.

WORK PIECES FOR MACHINE ACCEPTANCE
Both, pre-acceptance and final acceptance tests are carried out with either an EMAG test component to VDI standard, or the work piece specified by the customer's order.
The customer will supply an agreed number of work pieces for machining process optimisation and pre-acceptance tests. These work pieces must be from a single batch and of a quality that corresponds with the relevant manufacturing drawing.
For preparatory work and preliminary machine acceptance the customer will, at the agreed time, supply EMAG free of charge with a sufficient number of sample work pieces of a quality that corresponds with the relevant manufacturing drawing. Any delay in the supply of these work pieces will result in the machine delivery being delayed and the starting date of a possible delayed machine delivery penalty being deferred accordingly.

FINAL ACCEPTANCE AND FINAL PAYMENT
Final acceptance tests are carried out as soon as the machine has been installed and commissioned.
For this purpose, the buyer has to provide a sufficient number of work pieces, personnel and capacity to measure and evaluate the acceptance components.
Once the machine is taken into production and the agreed component quality and cycle time have been achieved, final acceptance is considered to have taken place.
Where final acceptance of the machine is not possible owing to reasons beyond EMAG's control (missing acceptance components, etc.), final contractual payment falls due immediately.

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TECHNICAL AVAILABILITY
Our confirmation of the technical availability according to VD I 3423 is based on following conditions:
Exclusively qualified operating personnel, monthly availability protocols addressed to our service manager, demonstrable preventive maintenance is carried out according to our machine documentation, connection to our tele service system if requested by our service department.
In case of non-compliance of these conditions, our commitment expires.

TECHNICAL DEVELOPMENT
In the interest of technical progress, we reserve ourselves the right to make design changes which are due to continuous product development.

RESERVATION CLAUSE
For offers within the European Union (without Germany):
„Notification of Intra-Community delivery according to Art. 22 (10) EC Dual-Use Regulation:
The following purchased goods are subject to export controls. For the export of these goods from the Community an export license is necessary."
For offers outside the European Union:
It is agreed, that the legally binding conclusion of this offer is subject to the suspensive condition that the competent authority grant the necessary export license.

(1) The Importer/Buyer shall not sell, export or re-export, directly or indirectly, to the Russian Federation or to Belarus, or for use in the Russian Federation or in Belarus, any goods supplied under or in connection with this Agreement that fall under the scope of (a) Article 12g of Council Regulation (EU) No 833/2014, and/or (b) Article 8 of the Council Regulation (EU) No 765/2006.

(2) The Importer/Buyer shall undertake its best efforts to ensure that the purpose of paragraph (1) is not frus-trated by any third parties further down the commercial chain, including by possible resellers.

(3) The Importer/Buyer shall set up and maintain an adequate monitoring mechanism to detect conduct by any third parties further down the commercial chain, including by possible resellers, that would frustrate the purpose of paragraph (1).

(4) Any violation of paragraphs (1), (2) or (3) shall constitute a material breach of an essential element of this Agreement, and EMAG shall be entitled to seek appropriate remedies, including, but not limited to: (i) termina-tion of this and any other affected Agreement; and (ii) a penalty of 10% (ten percent) of the total value of this and any other affected Agreement, or of the price of the all affected goods and services exported, whichever is higher.

(5) The Importer/Buyer shall immediately inform EMAG about any problems in applying paragraphs (1), (2) or (3), including any relevant activities by third parties that could frustrate the purpose of paragraph (1). The Im-porter/Buyer shall make available to EMAG information concerning compliance with the obligations under par-agraph (1), (2) and (3) within two weeks of the simple request of such information.

ADDITIONAL PROVISIONS
In addition and subordinated to this quotation our General Conditions for Sales and Deliveries for International Business shall apply.

EMAG GENERAL CONDITIONS FOR SALES AND DELIVERIES International Business

https://www.emag.com/de/iot-services.html
https://www.emag.com/media-recommendation/

Price does not include options (*)

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This document was created electronically and is valid without signature.

Sincerely,

EMAG Maschinenfabrik GmbH

Hartmut von Fugler
Managing Director

Do-Youn Koo
Area Sales Manager

Attachments

EMAG-General conditions for sales and deliveries (UN)
Layout
Schematic of machining
Timestudy
Quality commitments

Price does not include options (*)

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